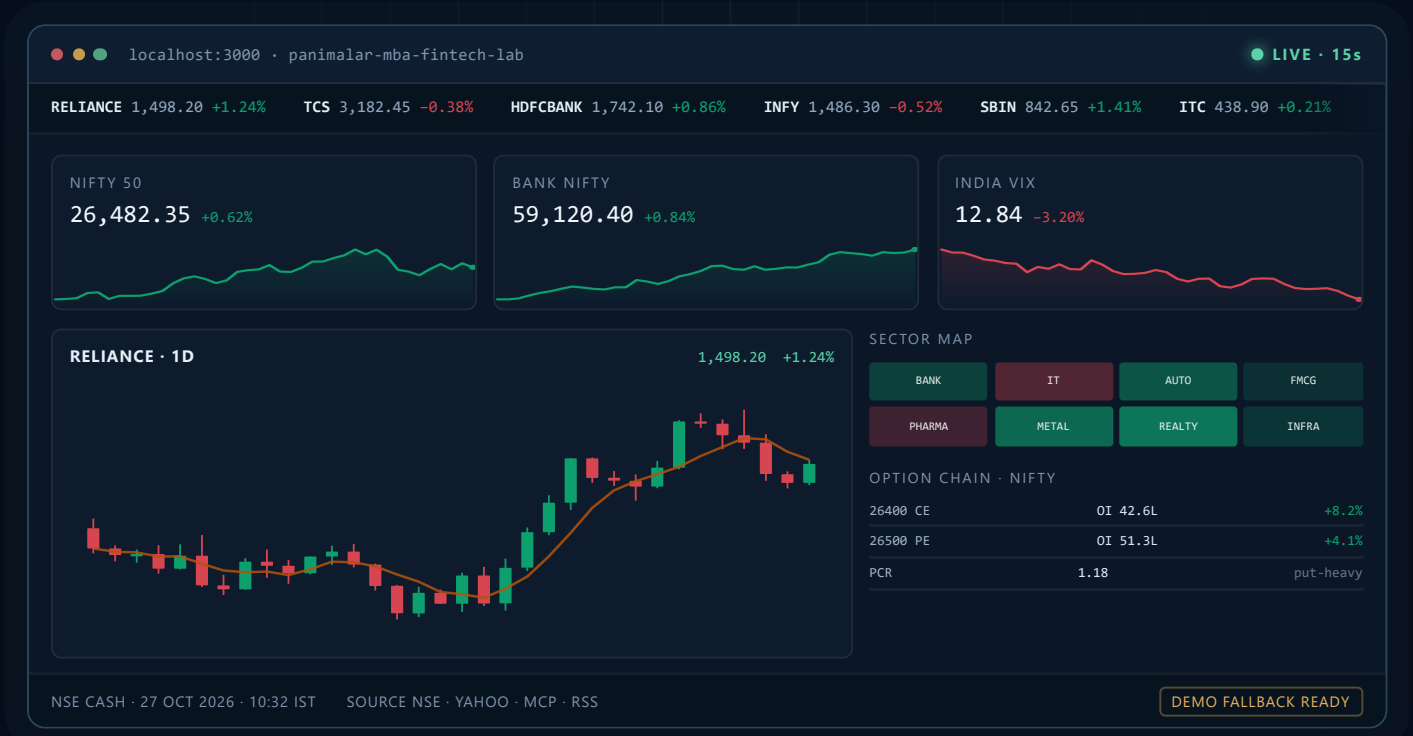
PANIMALAR ENGINEERING COLLEGE
Department of Management Studies · Chennai

PROSPECTUS
2026–27

MBA · FINANCE · ANALYTICS · FINTECH

The Fintech Lab Terminal

Live NSE market intelligence, engineered for the classroom. **Twelve research modules, three honest data states**, and a screen that never goes blank — so a case discussion can start with a chart, not a slide.



01
Read the market
Indices, breadth, movers, sector map — the state of the session.

02
Build the analysis
Candles, ratios, option chain, institutional flows.

03
Defend the decision
A note, a hedge, a portfolio review — graded on evidence.

12	12	3	0	15s
RESEARCH MODULES	API ENDPOINTS	HONEST DATA STATES	API KEYS	DEFAULT REFRESH

LIVE · DELAYED · DEMO

INSIDE FEATURES · REAL-WORLD USE CASES
12-WEEK CURRICULUM · LAB SETUP & API

01

THE PROPOSITION

One screen. Twelve ways to learn finance.

Most MBA finance courses end with a valuation spreadsheet built on a case that is three years old by the time it is graded. This terminal puts the **current market session** on the projector — the index, the option chain, the institutional flows — and labels every number with where it came from and how old it is.

It is built for a 40-student lab and a dim projector: dense, dark, readable from the back row, no login and no API keys. Students open a browser and immediately start asking the questions an analyst asks.

DESIGNED FOR

Investment analysis
Valuation, screening, research notes

Derivatives & risk
Option chain, volatility, hedging

Business analytics
Data quality, dashboards, storytelling

FinTech product
APIs, latency, graceful failure

THE TWELVE MODULES

Every panel is a teaching object, not decoration.

Ticker tape
Watchlist and movers scrolling with LTP and % change.

Index cards
NIFTY 50, Bank NIFTY, IT, Midcap, Next 50 and India VIX.

Market breadth
Advance / unchanged / decline rail for the NIFTY 50.

Watchlist
Editable, sortable, saved per browser for each student.

Price chart
Intraday area and 1W–1Y candlesticks with volume.

Movers
Top gainers, losers and volume toppers in the NIFTY 50.

FII / DII flows
Institutional cash-market buy, sell and net in ₹ crore.

Sector heat-map
Twelve sectoral indices, colour-scaled by % change.

Option chain
ATM ± 5 strikes: OI, OI change, LTP, IV and PCR.

Market news
Latest Economic Times headlines, timestamped.

Stock detail
52-week range, P/E, ROE, yield — with explainers.

MarketLens
One-click bridge to NSE's official screener and valuation modules.

12	12	3	0	15s	40
RESEARCH MODULES	API ENDPOINTS	DATA STATES	API KEYS	DEFAULT REFRESH	STUDENTS PER CACHE

02 Why it works — architecture and data truth

03–05 Feature tour — pulse, research, derivatives

06–07 Seven real-world use cases

08–09 12-week curriculum, setup, API, ethics

	A FREE FINANCE SITE	A VIDEO CASE	THIS TERMINAL
LIVE MARKET	Delayed, ad-supported, no context	No — recorded months ago	**Current session, cached for 15 seconds**
PROVENANCE	Rarely stated	Fixed narrative	**Status, source and as-of on every card**
DERIVATIVES	Buried, or behind a paywall	Not available	**Option chain with OI, IV and PCR**
EXTEND IT	Closed product	Not possible	**12 documented JSON endpoints**
A BAD DATA DAY	Error page	Not applicable	**Labelled DEMO feed, class continues**

02

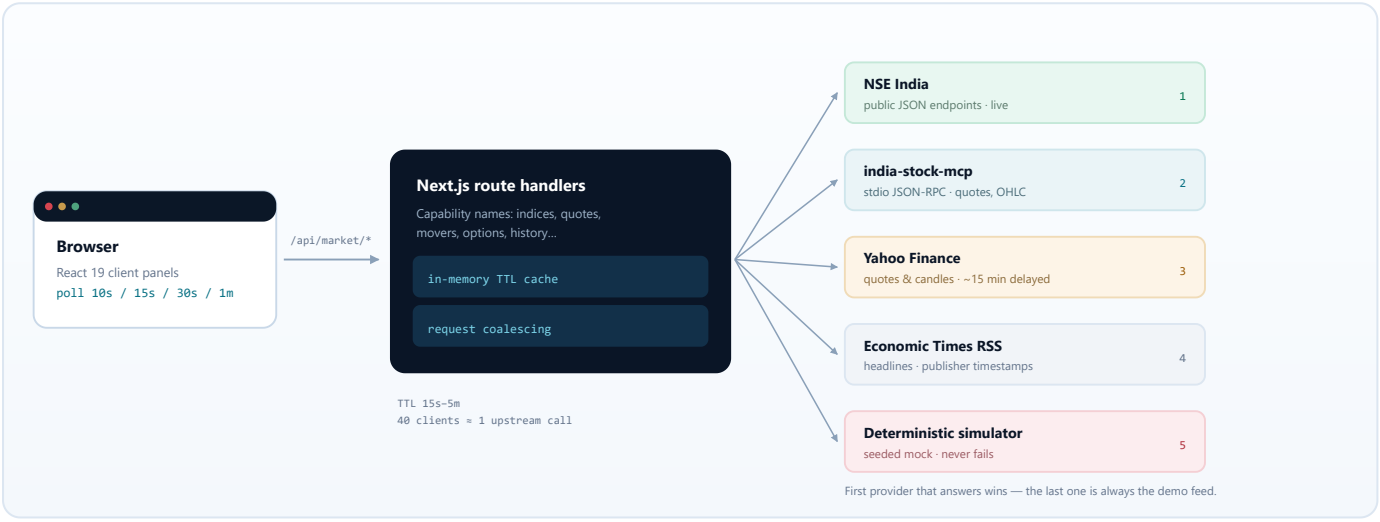
UNDER THE HOOD

Every number is labelled. That is the lesson.

A dashboard is only as credible as its provenance. Each response carries a **status**, a **source**, an **as-of timestamp** and an optional warning. When NSE is unreachable, the request falls through to the MCP server, then Yahoo Finance, then a deterministic simulator — so the class never stares at a spinner, and never mistakes a simulation for the exchange.

The same discipline applies to an analyst's model: a number without a source and a date is an opinion.

```
// every market response
{
  "data": [ … ],
  "status": "live",
  "source": "nse:indices",
  "asOf": "2026-10-27T10:32:11+05:30",
  "warning": null
}
```



Never blank

Every capability ends in the simulator. A dropped feed changes the label, not the lesson plan.

Label the truth

LIVE, DELAYED or DEMO on every panel — plus the actual source and timestamp.

Cache once, serve forty

TTLs from 15 seconds to 5 minutes; hidden tabs drop to one request per minute.

- **LIVE** Straight from the exchange endpoint. Freshest state.
- **DELAYED** Yahoo or MCP fallback, typically 15 minutes behind.
- **DEMO** Seeded simulation; the banner explains why it appeared.

WHY AN MBA SHOULD CARE

Analysts spend more time validating data than modelling. Asking "where did this number come from, and how old is it?" is a professional habit — and this terminal makes that habit visible on every card.

FALLBACK ORDER, BY CAPABILITY

The first provider that answers wins

CAPABILITY	FIRST CHOICE	THEN	TTL
Indices & quotes	NSE public JSON	MCP → Yahoo → demo	15 s
Candles & history	Yahoo Finance	MCP → demo	60 s
Option chain	NSE option chain	MCP → demo	45–60 s
Institutional flows	NSE cash market	Demo only (latest session)	5 min
Headlines	Economic Times RSS	Demo only	5 min

Citable in a graded memo: LIVE with its timestamp · DELAYED stated as indicative · DEMO for method only, never as market evidence.

Browser poll → Route handler → TTL cache → Provider chain → Data envelope

Same path for all twelve modules.

Lab drill: break the feed

10 minutes · first session

1. Disconnect the lab network and reload the dashboard.
2. Read the banner: which panels switched to DEMO?
3. Open `/api/health` and name the failing provider.
4. Reconnect, refresh, and note which panel recovered first.

03

FEATURE TOUR · I

The market at a glance — and what each glance hides.

RELIANCE 1,498.20 +1.24% TCS 3,182.45 −0.38% HDFCBANK 1,742.10 +0.86% as of 10:32

NIFTY 50 Benchmark
26,482.35 +0.62%

L 26,318 H 26,540

BANK NIFTY Banking
59,120.40 +0.84%

L 58,640 H 59,285

NIFTY IT Technology
39,850.15 −0.41%

L 39,702 H 40,118

MIDCAP 100 Mid caps
62,340.80 +1.12%

L 61,702 H 62,410

NEXT 50 Large caps
74,210.05 +0.35%

L 73,942 H 74,368

INDIA VIX Volatility
12.84 −3.20%

L 12.61 H 13.42

MARKET BREADTH · NIFTY 50

34 / 3 / 13



34 advancing · 3 unchanged · 13 declining

GAINERS · LOSERS · VOLUME	% change
REALTY · DLF	+3.42%
METAL · TATASTEEL	+2.86%
PSU BANK · SBIN	+1.41%
LOSERS	
IT · TCS	−0.38%
PHARMA · SUNPHARMA	−0.62%
VOLUME TOPPER	
YESBANK	2.4× avg

WHAT TO LOOK AT, AND WHY

1. **The tape is relative**
   Students compare % change, not price. A ₹12,000 stock and a ₹400 stock are only comparable as percentages.
2. **Six windows, one market**
   Large cap vs mid cap vs IT vs volatility. Rotation begins as a relative number on a card.
3. **Sparklines show character**
   Direction, slope and volatility in one glance. VIX falling while the index rises is the classic risk-on signature.
4. **Breadth beats the headline**
   An index can rise while most constituents fall. The advance/decline rail exposes a narrow rally.
5. **Movers separate news from noise**
   Gainers and losers show reaction; volume toppers show conviction. Ask which move has volume behind it.
6. **Session status is context**
   Pre-open, live and post-close change how a quote should be read. Every panel carries its own clock.

SESSION STATUS

● OPEN

10:32 IST · Tuesday

Continuous trading since 09:15. Closes 15:30. Quotes are cached for 15 seconds.

pre-open 09:00 · auction 09:08 · close 15:30

ANATOMY OF A CARD

1	**Badge**	LIVE, DELAYED or DEMO — read it before the number.
2	**Value**	Last cached tick, with its change against the previous close.
3	**Sparkline**	Session path on the day's own scale, so shape survives.
4	**L & H**	Session low and high — not the 52-week range.

SIXTY-SECOND CHECK

every card

1. Is the move broad, or three stocks in a trench coat?
2. Which sector is doing the work?
3. What would have to be true for it to reverse?

A TRADING DAY IN IST — THE CLOCK STUDENTS MUST INTERNALISE

Monday to Friday

09:00	09:08	09:15	12:00	15:30	15:40
Pre-open session	Call auction	Continuous trading	Mid-session	Close	Post-close

A rising index, a falling majority
Breadth divergence: a handful of heavyweights can lift the index while most stocks decline. **Ask:** how many stocks are advancing?

One sector in disguise
Concentration: a "broad" index can be a single sector story. **Ask:** which sector drove the move?

A one-day move is not a trend
Timeframe: the same +2% is noise on a 1Y chart and a breakout on a 1D chart. **Ask:** over what horizon?

Panel coverage · NIFTY 50 universe + 60 liquid symbols

Refresh 10s – 1m

04

FEATURE TOUR · II

From a ticker to an investment note.

Watchlist 12 symbols

DELAYED MCP → Yahoo · 15s

COMPANY	LTP	CHANGE	CHANGE %	DAY RANGE	VOLUME	P/E
RELIANCE Reliance Industries	1,498.20	+18.35	+1.24%	1,472–1,502	62.4L	24.8
TCS Tata Consultancy	3,182.45	−12.10	−0.38%	3,168–3,210	12.1L	26.9
HDFCBANK HDFC Bank	1,742.10	+14.90	+0.86%	1,721–1,748	38.7L	19.4
INFY Infosys	1,486.30	−7.80	−0.52%	1,479–1,501	21.3L	22.1
ICICIBANK ICICI Bank	1,284.55	+9.20	+0.72%	1,271–1,289	31.8L	18.2
SBIN State Bank of India	842.65	+11.75	+1.41%	828–845	54.2L	11.6
BHARTIARTL Bharti Airtel	1,638.40	−4.10	−0.25%	1,630–1,652	9.8L	32.4
MARUTI Maruti Suzuki	12,864.00	+137.50	+1.08%	12,702–12,910	4.2L	27.5

RELIANCE 1,498.20 +1.24%

1D 1W 1M 6M 1Y



OHLC candlesticks · volume · 5-period moving average

DELAYED Source: MCP / Yahoo

READING THE CHART

Structure before prediction

Higher lows from mid-session, then a break above the prior swing high. Students describe what happened before forecasting.

Volume confirms the move

The breakout bar carries roughly 2.5× average volume. A price move without volume is a hypothesis, not evidence.

The y-axis is a choice

Switch from 1D area to 1Y candles and the same stock tells a different story. Timeframe is an analytical decision.

RATIO EXPLAINERS

hover any metric

Trailing P/E sector 20.1×	24.8×
Forward P/E	22.1×
Return on equity	18.4%
Dividend yield	0.6%
52-week range	1,214–1,512

Each ratio carries a one-line definition and a benchmark, so the number becomes an argument students can defend.

MARKETLENS DRAWER

NSE portal

One keystroke opens the official NSE screener for deeper fundamental work.

`P/E < 20` `ROE > 15%` `D/E < 1` `Promoter pledge = 0` `Market cap > ₹50k cr`

Screen 2,000+ listed equities Open ↗

FROM SCREEN TO NOTE

workflow

Search → Watch → Compare → Explain → Write

Press **/** to search 70+ symbols, **W** to add or remove, **Enter** for the detail card. The watchlist persists per browser, so each student keeps a personal universe.

localStorage sortable keyboard-first

05

FEATURE TOUR · III

Where price meets positioning.

NIFTY option chain

weekly

26,482.35 +0.62%

spot · ATM ± 5 strikes

OI (L)	CHG	LTP	IV	STRIKE	IV	LTP	CHG	OI (L)
38.4	+6.2	312.5	11.8	26,200	12.9	38.2	−18.4	72.4
40.1	+4.8	268.0	11.6	26,250	13.1	62.4	−14.2	66.1
44.2	+9.1	221.5	11.4	26,300	12.8	96.8	−11.6	58.3
47.8	+7.4	176.0	11.2	26,350	12.6	142.5	−8.9	53.0
52.3	+11.2	134.5	11.0	26,400	12.4	198.2	−6.2	49.2
58.9	+14.6	98.0	10.9	26,450	12.3	262.4	−4.1	46.0
49.8	+4.1	68.5	10.8	26,500	12.2	338.9	−2.8	60.4
43.7	−2.6	46.2	10.9	26,550	12.4	412.6	−1.9	42.2
36.2	−5.3	30.8	11.1	26,600	12.7	489.5	−1.2	37.8

Calls · OI change in lakh contracts

Puts · 9 strikes

POSITIONING GAUGE

LIVE

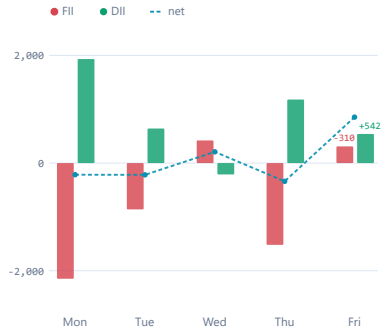


Put OI vs call OI	485L / 411L
ATM implied volatility	10.8%
Max pain	26,500

A PCR above 1.0 means more puts than calls are outstanding. It describes positioning — not a buy or sell signal.

FII / DII CASH FLOWS

₹ crore



Net: **−₹216 cr** on the session. Institutions are not a single actor — domestic funds can absorb foreign selling.

5-SESSION FII	5-SESSION DII	COMBINED
−3,797	+4,083	+286

SECTOR HEAT-MAP

12 indices

BANK +0.84%	IT −0.41%	AUTO +1.32%
FMCG +0.18%	PHARMA −0.62%	METAL +1.94%
REALTY +2.31%	ENERGY +0.44%	PSU BANK +1.08%
FIN SRV +0.72%	MEDIA −0.25%	INFRA +0.51%

Breadth within sectors matters: a green tile can hide a narrow, two-stock rally.

MARKET NEWS

RSS

Banks lead Q2 earnings as NIM recovery beats estimates
10:24 IST · Markets

FII selling slows; domestic funds step in for third session
09:58 IST · Economy

Auto index at record high on festive demand, easing input costs
09:41 IST · Companies

RBI holds repo rate; commentary signals an extended pause
09:22 IST · Economy

IT guidance trimmed as deal cycles lengthen in BFSI
09:18 IST · Companies

Headlines explain the narrative; the tape and the flows test whether the narrative is priced.

1. **Locate spot** — 26,482 → nearest strike 26,500.
2. **Find the walls** — Highest put OI: 26,200 (72L); call wall: 26,450 (59L).
3. **Read the price of risk** — PCR 1.18, ATM IV 10.8%, VIX 12.8.
4. **Test a decision** — Is the put cheap enough to insure the portfolio?

THREE TRAPS TO TEACH

Open interest is not volume. A large OI can be old positioning that never trades today.

PCR is a ratio, not a signal. It needs price context, OI change and expiry to mean anything.

Implied volatility is the price of uncertainty. Rising IV makes every option more expensive, including the hedge.

06

REAL-WORLD USE CASES · I

Three desks, three deliverables.

Each case runs for two lab sessions and ends with a document a real desk would accept. The terminal supplies the evidence; the student supplies the judgement.

CASE 01

Equity research analyst

Investment Analysis & Portfolio Management
2 sessions · pairs

relative valuation

Scenario. You cover Indian IT. NIFTY IT is down 0.4% while the benchmark is up 0.6%. Write a two-page initiation note on Infosys against TCS, with a clear recommendation and a stated data cutoff.

1 Screen the IT universe
2 Pull quote, 52W, P/E, ROE
3 Compare chart behaviour
4 Write with as-of stamps

Learn: multiples vs peers, earnings quality, relative strength, data hygiene, and how to write a recommendation that survives a viva.

Watch for: comparing P/E without adjusting for growth, and a note with no data cutoff.

DATA `watchlist` `stock detail` `1Y chart` `MarketLens`

P/E VS SECTOR · TRAILING



TCS 14% above the sector, INFY 6% below — argue which is earned.

Deliverable
2-page note + 5-minute defence · P/E vs sector shown, every figure dated.

CASE 02

Portfolio manager

Security Analysis & Portfolio Management
3 sessions · groups of 4

construction

Scenario. Build a ₹10 lakh core-satellite portfolio of eight stocks. Cap any sector at 25%, benchmark against the NIFTY 50, and review it every lab session for four weeks.

1 Define the mandate
2 Set weights & caps
3 Track daily % and breadth
4 Write the rebalance memo

Learn: diversification, concentration limits, tracking difference, performance attribution and the discipline of not rewriting the thesis after the fact.

Watch for: sector caps breached after a winner runs, and benchmark switching mid-stream.

DATA `index cards` `breadth` `movers` `sector map`



Deliverable
Portfolio sheet + weekly review · compare against NIFTY 50, explain every change.

CASE 03

Derivatives & risk

Derivatives Management
2 sessions · groups of 3

hedging

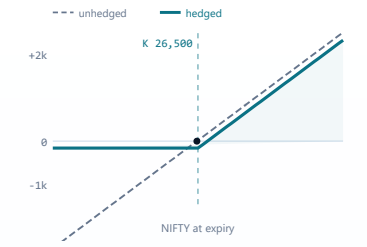
Scenario. Protect the portfolio from Case 02 with a NIFTY protective put. India VIX is 12.8. Choose the strike and expiry, price the insurance, and decide whether the hedge is worth it.

1 Read the chain
2 Pick strike & expiry
3 Compute premium cost
4 Draw the payoff

Learn: moneyness, open interest as positioning, implied vs realised volatility, the cost of insurance and basis risk between a stock portfolio and an index hedge.

Watch for: a hedge notional that does not match the portfolio, and a payoff that ignores the premium.

DATA `option chain` `India VIX` `PCR` `payoff`



MAX LOSS	BREAKEVEN	COST
180	26,680	0.7%

Deliverable
Payoff diagram + one-page hedge memo · premium, breakeven, worst case.

> The terminal is not the answer. It is the evidence.
>
> LAB PRINCIPLE — EVERY DELIVERABLE IS A DECISION, NOT A DESCRIPTION

07

REAL-WORLD USE CASES · II

Four shorter briefs. Same standard of proof.

CASE 04

explain

Macro & institutional flows

45 MIN · PAIRS · 150-WORD MEMO

The brief. FIIs sold ₹2,148 crore and DIIs bought ₹1,932 crore, yet the NIFTY rose 0.62%. Explain who absorbed the supply and why the index still moved up.

−2,149	+1,932	−216	12.84
FII ₹ CR	DII ₹ CR	NET ₹ CR	INDIA VIX

WHO ABSORBED THE SALE? ₹ CRORE



FII sold 2,149 · DII bought 1,932 · others +216

1. Read the flow panel and compute the net.
2. Check breadth, VIX and the sector map for confirmation.
3. Write 150 words: flow, absorption, narrative, one risk.

DATA `fii-dii` `breadth` `heat-map` `news`

> **Watch for:** treating one session as a trend, and calling a net outflow bearish on its own.

Deliverable: 150-word memo + one named risk.

liquidity · flows vs fundamentals · narrative discipline

CASE 05

allocate

Sector rotation strategy

60 MIN · GROUPS OF 3 · ROTATION NOTE

The brief. Plot the twelve sectoral indices on a relative-strength / earnings-momentum map, then propose an overweight and an underweight for the next quarter.

DEFENSIVE · FADING
PHARMA −0.6 · FMCG +0.2 · IT −0.4

MOMENTUM LEADERS
METAL +1.9 · REALTY +2.3 · AUTO +1.3 · PSU BK +1.1

UNDERWEIGHT
MEDIA −0.3

IMPROVING · UNCONFIRMED
BANK +0.8 · ENERGY +0.4 · INFRA +0.5

← weaker relative strength · earnings momentum ↑ · stronger →

DATA `sector map` `index cards` `movers`

> **Watch for:** rotating on one week of price, and ignoring valuation when chasing momentum.

Deliverable: one-page rotation note with two positions.

business cycle · relative strength · position sizing

CASE 06

design

FinTech product manager

45 MIN · PAIRS · FAILURE MAP

The brief. Trace one quote from provider to screen. List three failure modes, then redesign the badge and fallback copy so a student never mistakes a simulation for the exchange.

NSE → MCP → Yahoo → DEMO

TTL 15s · coalescing · status / source / asOf

WHERE QUOTES ACTUALLY CAME FROM LAST 30 DAYS



NSE 74% · MCP 18% · DEMO 8%

1. Map the fallback chain and its labels.
2. Choose a TTL and justify the freshness trade-off.
3. Write the degraded-state copy for a projector.

DATA `/api/health` `badges` `TTL`

> **Watch for:** a DEMO badge that reads like LIVE, and a retry loop that hides the outage.

Deliverable: failure-mode map + revised copy.

SLA · graceful failure · trust design

CASE 07

build

Business analyst, BFSI

90 MIN · INDIVIDUAL · DASHBOARD

The brief. Pull four endpoints into Excel or Power BI, document the schema, and publish a one-screen classroom dashboard with four KPIs and a written caveat.

`/api/market/quotes` → CSV / JSON → Power BI → 4 KPIs

26,482	+0.62%	34/13	−216
NIFTY	1D CHANGE	BREADTH	NET FLOWS

1. Write the data dictionary: field, type, source, refresh.
2. Handle nulls, stale rows and unknown symbols.
3. Present the chart that changes the decision, not the prettiest one.

DATA `quotes` `history` `options` `fii-dii`

> **Watch for:** a dashboard that refreshes without a timestamp, and nulls silently plotted as zero.

Deliverable: working dashboard + data dictionary.

ETL · data quality · visual storytelling

WHERE THIS LEADS

Placement-facing skills

equity research associate · portfolio / wealth analyst · risk & derivatives analyst · fintech product analyst · business analyst (BFSI)

08

THE 12-WEEK LAB

Twelve weeks. One terminal. Seven artefacts.

W1–2
Orientation
Market structure, data provenance, badges

W3–4
Research
Multiples, peers, the initiation note

W5–6
Price behaviour
Candles, volume, timeframe choice

W7–8
Portfolio
Weights, caps, benchmarking

W9–10
Derivatives
Option chain, volatility, hedging

W11–12
Product & capstone
Fallbacks, API, final defence

WEEK	MODULE	TERMINAL MODULES USED	DELIVERABLE	BLOOM
1–2	**Market microstructure & data provenance**	Ticker, index cards, session status, badges, `/api/health`	Lab journal: decode one live session	Understand
3–4	**Equity research & relative valuation**	Watchlist, stock detail, ratio explainers, MarketLens	Two-page initiation note	Apply
5–6	**Price behaviour & technical evidence**	Candles, volume, 1D–1Y ranges, movers	Chart-based hypothesis memo	Apply
7–8	**Portfolio construction & benchmarking**	Index cards, breadth, movers, sector map	₹10 lakh, eight-stock portfolio	Analyse
9–10	**Derivatives, volatility & hedging**	Option chain, PCR, India VIX, payoff	Protective-put hedge memo	Analyse
11	**Macro flows & sector rotation**	FII/DII, heat-map, news feed	"Explain yesterday" brief	Evaluate
12	**FinTech product & capstone**	All modules plus the API surface	10-minute pitch + viva	Create

0–10 MIN
The question
One market event, stated as a decision to make.

10–25
Demo
One panel, one workflow, narrated on the projector.

25–65
Build
Students produce the artefact in pairs or groups.

65–80
Critique
Swap, challenge the evidence, mark one gap.

80–90
Journal
Two lines: what changed in your thinking, and why.

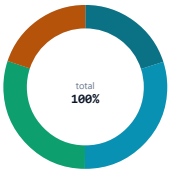
W2
Journal baseline
One live session decoded, every number sourced.

W6
Mid-term review
Initiation note and chart memo, peer-marked.

W10
Hedge review
Payoff, premium and basis risk defended in class.

W12
Capstone viva
One page, four modules, one decision held up to challenge.

ASSESSMENT WEIGHT



- **20%** Lab journal
- **30%** Assignments
- **30%** Capstone
- **20%** Viva

Continuous internal assessment; the journal is marked weekly, not at the end.



WHAT "GOOD" LOOKS LIKE

EVIDENCE	Every claim traces to a panel, a source and a timestamp.
REASONING	Alternatives considered; the counter-case stated, not hidden.
DECISION	A position with a horizon, a risk and an exit.
COMMUNICATION	One page, one chart, one recommendation a manager can act on.

CAPSTONE BRIEF

Choose one NSE stock. Build a one-page investment case using at least four terminal modules. Declare the status, source and as-of for every data point. Then defend one hedge against it.

4 modules · 1 page · 10 min

Evidence trail · one screenshot per module, each carrying the badge and the as-of time.
Submit · PDF note + link to your dashboard.
Viva · 5-minute defence.

Before week 1
Form groups of four, pick a sector and read the project README.

Hardware
One laptop per pair; one projector for the demo and critique.

Software
Students need a browser only. Node.js 22+ runs on the lab host.

Cost
No licences, no API keys, no market-data subscription.

09

OPERATE IT

Two commands. Then start asking questions.

```
# Node.js 22+ · no API keys
npm install  # installs india-stock-mcp too
npm run dev  # http://localhost:3000

# lab server, reachable from lab PCs
npm run build && npm start # 0.0.0.0:3000
```

GET /api/health

Per-provider status, timings and the MCP tool list.

CHECK

KEYBOARD-FIRST, BECAUSE A PROJECTOR HAS NO MOUSE

/ Search	? Shortcuts	R Refresh	Space Pause
[ ] Rate	↑ ↓ Step	Enter Detail	W Watch
1–5 Range	C Chart	V Volume	F Full scr
Z Zoom	T Theme	L MarketLens	M Dock tape

Pop-out any panel into a second window for a dual-monitor lab; symbol changes broadcast between windows.

THE API STUDENTS CAN BUILD ON

ENDPOINT	RETURNS	TYPICAL TTL
`GET /api/market/indices`	Index quotes with day range and advance/decline	15 s
`GET /api/market/quotes?symbols=`	Equity quotes for a comma-separated watchlist	15 s
`GET /api/market/options?symbol=NIFTY`	Trimmed option chain around ATM	45–60 s
`GET /api/market/fii-dii`	Institutional cash-market flows in ₹ crore	5 min
`GET /api/market/history?symbol=&range=`	OHLCV candles for 1D, 1W, 1M, 6M or 1Y	60 s
`GET /api/market/news?limit=`	Publisher headlines with timestamps	5 min

KNOWN LIMITATIONS — TAUGHT, NOT HIDDEN

- NSE has no official public API; some endpoints return 403 on campus networks, hence the MCP and Yahoo fallbacks.
- Yahoo quotes are delayed by roughly 15 minutes — fine for teaching, not for trading.
- FII/DII publishes only the latest session through the public endpoint.
- Symbols change; unknown tickers render as “no quote” instead of disappearing.

DATA ETHICS & ATTRIBUTION

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Room
Projector at 1600–1920 px, F11 full screen, dark theme by default.

Network
One host machine; 40 browsers share the cache. No keys, no student installs.

Software
Node.js 22+ on the host; any modern browser on the lab PCs.

Offline
Pull the cable: every panel falls back to DEMO and the class continues.

IF SOMETHING LOOKS WRONG

Three faults students hit in week one.

A panel is blank
Open `/api/health`; the badge will say whether it is delayed or demo.

NSE returns 403
Expected on many campus networks. Quotes fall back to MCP, then Yahoo.

Unknown ticker
NSE renames symbols (TATAMOTORS → TMCV / TMPV). Edit the watchlist or `symbols.ts`.

Where it fits the syllabus

One terminal, five core courses

FM Financial Management · SAPM Security Analysis & Portfolio Management · DER Derivatives Management · BA Business Analytics · FINTECH Digital Banking & FinTech

Panimalar MBA Fintech Lab
Department of Management Studies · Panimalar Engineering College · Chennai

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